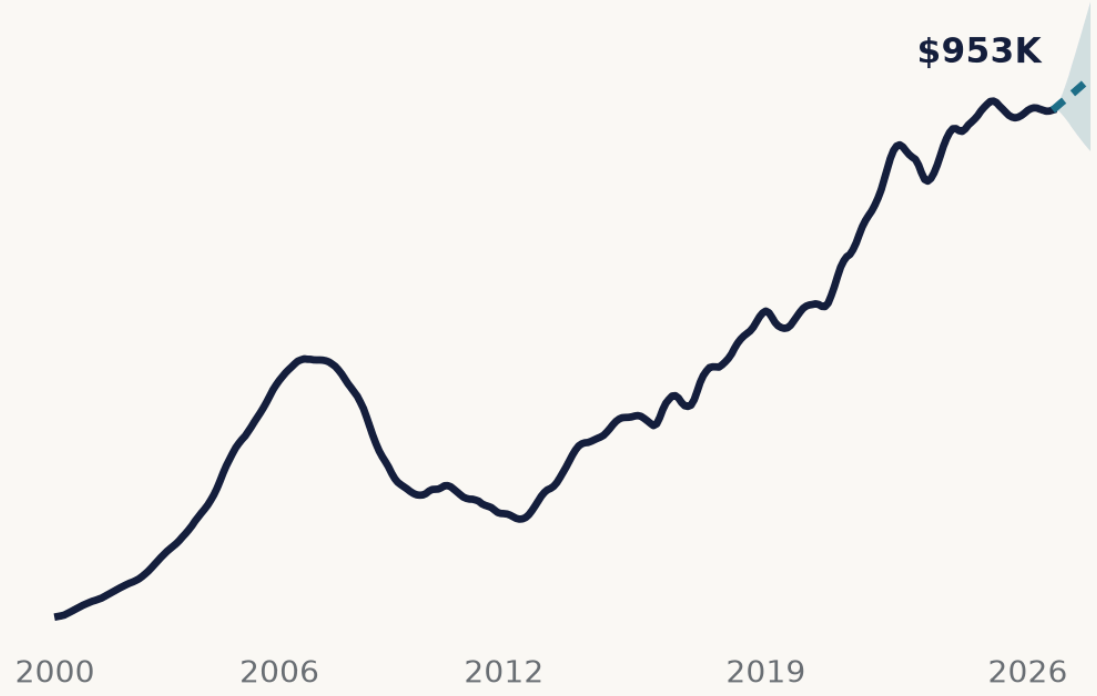
MGMTMSA 437 · Final project · October 2026

Forecasting LA Home Values

How well can we forecast LA home values up to a year ahead, and which works best: time-series models or AI?

Yang Liu · Jinwoo Roh
Yiu-Hon Lee · Thanapol Doungsaeng

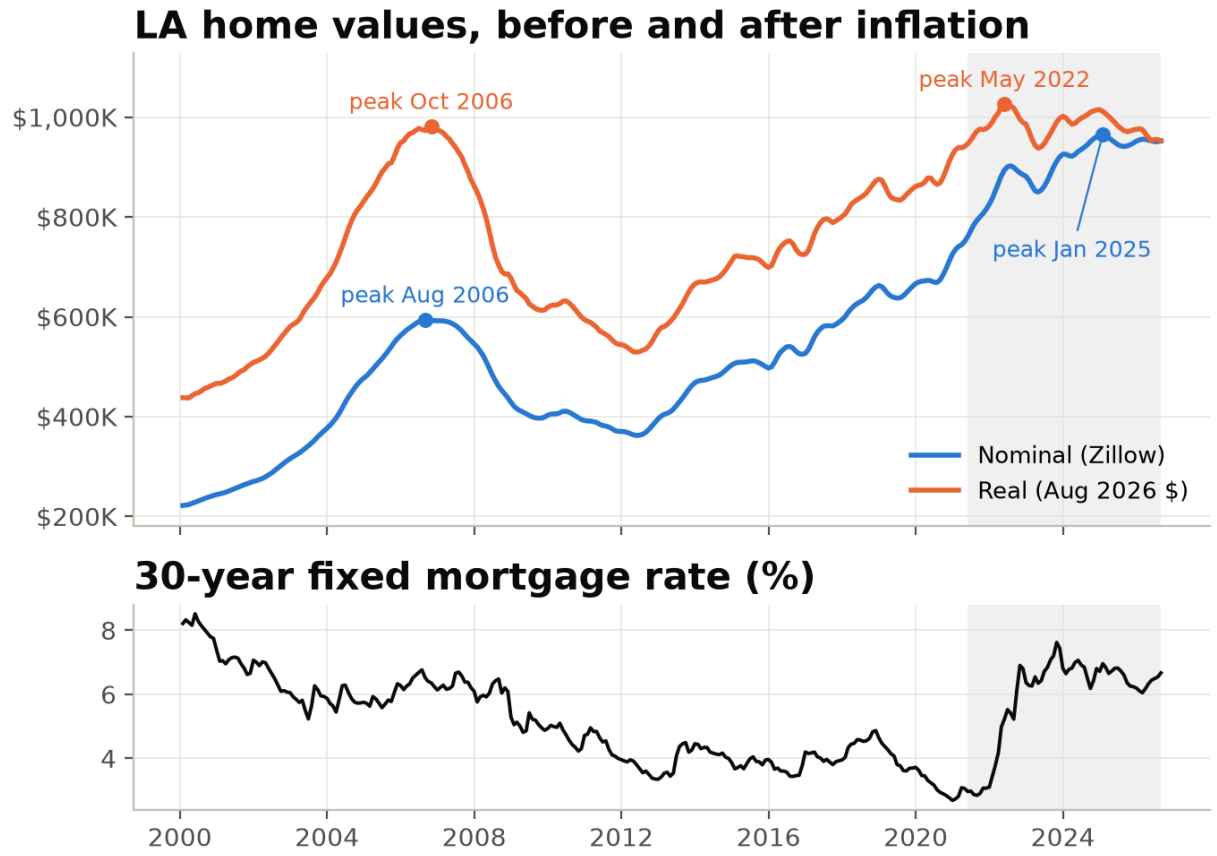
Monthly Zillow home values for the LA metro, January 2000 to August 2026, with mortgage rates and neighborhood data



Solid: actual · Dashed: our forecast to Aug 2027

Why LA, why now

After decades of growth, LA home values have stalled



$952,601

Typical home value, LA metro, August 2026

-1.3%

Below the January 2025 peak, before inflation

-7.1%

Below the May 2022 peak, after inflation

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

Data and design

We tested 14 models at 18 forecast dates

Data

Zillow ZHVI, LA metro
320 months, Jan 2000 to Aug 2026

499 LA neighborhoods
Zillow neighborhood index

Mortgage rate and payroll jobs
FRED and BLS, LA and Orange County

14 models + 3 AI

Baseline and classical
Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI
XGBoost · LightGBM · LSTM · GRU · Chronos

Outside information
ARIMAX ×2 · VAR · factor VAR

Plus 3 AI models (LLMs)
Claude · GPT · Gemini

How we tested

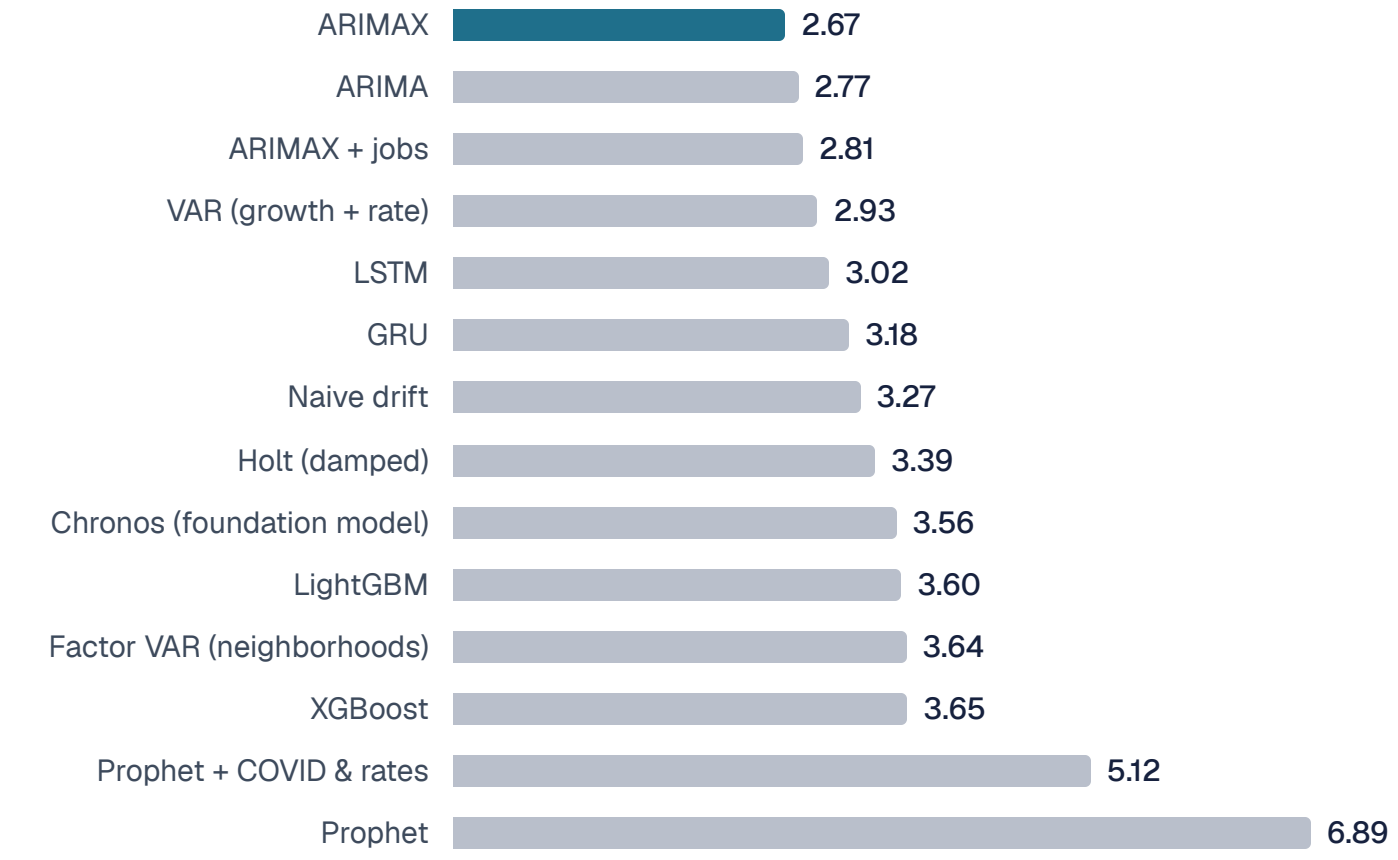
1. Rolling-origin test (main)
Repeated 12-month forecasts: update every 3 months, 18 start dates

2. One long forecast
Train on the first 80% (to April 2021), then forecast the next 64 months in one shot

How we score MAPE measures how far forecasts are from actual values, in percent. We average the absolute errors across months 1 to 12 and all 18 start dates. Lower is better.

Result 1 · Rolling-origin test

ARIMAX had the lowest average forecast error



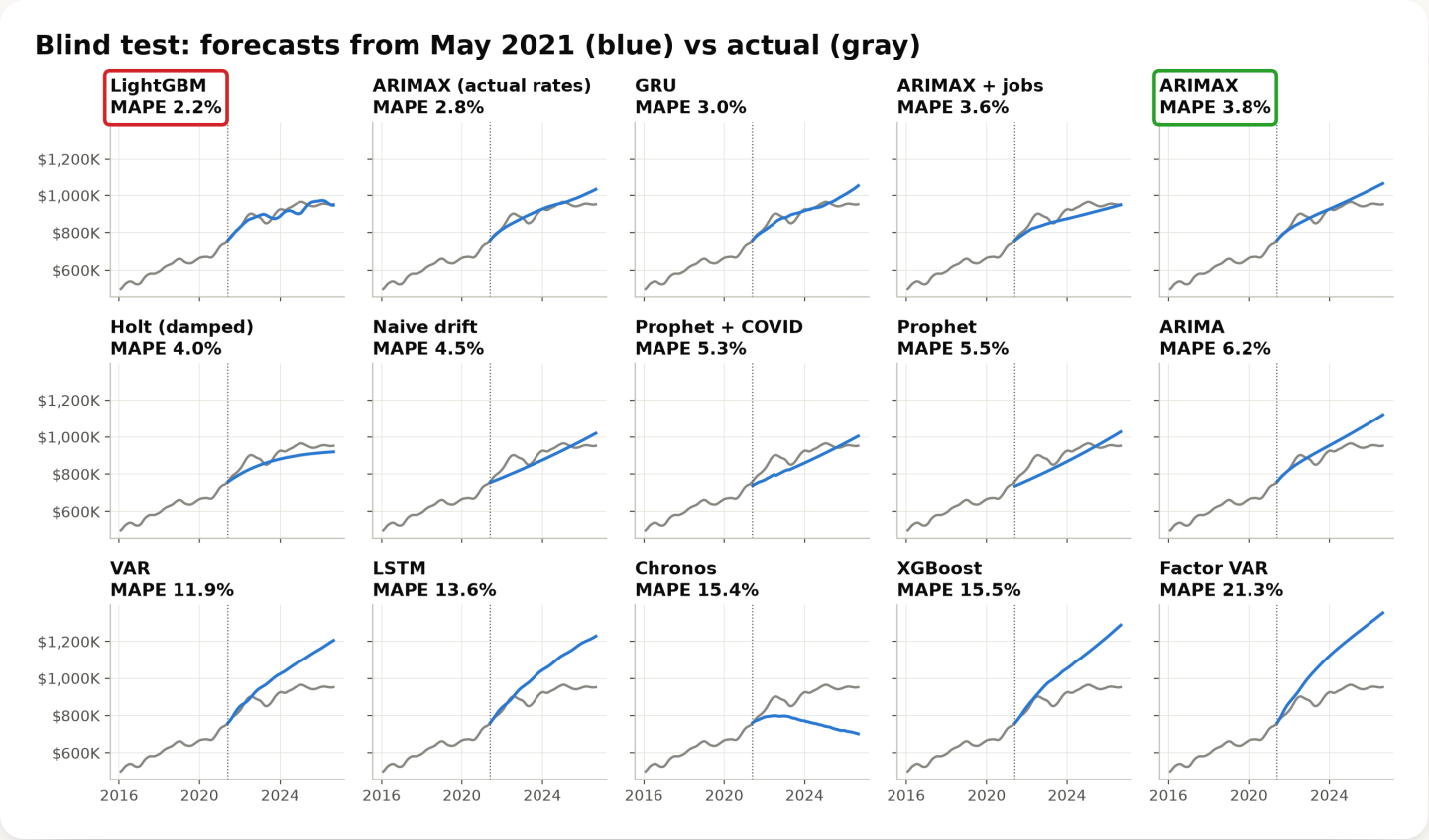
Lowest error (MAPE)

2.67%

average error across months 1 to 12. ARIMAX adds mortgage-rate information to ARIMA; plain ARIMA was close, at 2.77%. At month 12 alone: ARIMAX 5.0%, naive drift 4.9%.

Result 2 · One long forecast

LightGBM ranked first in one long test, but tenth in repeated 12-month tests

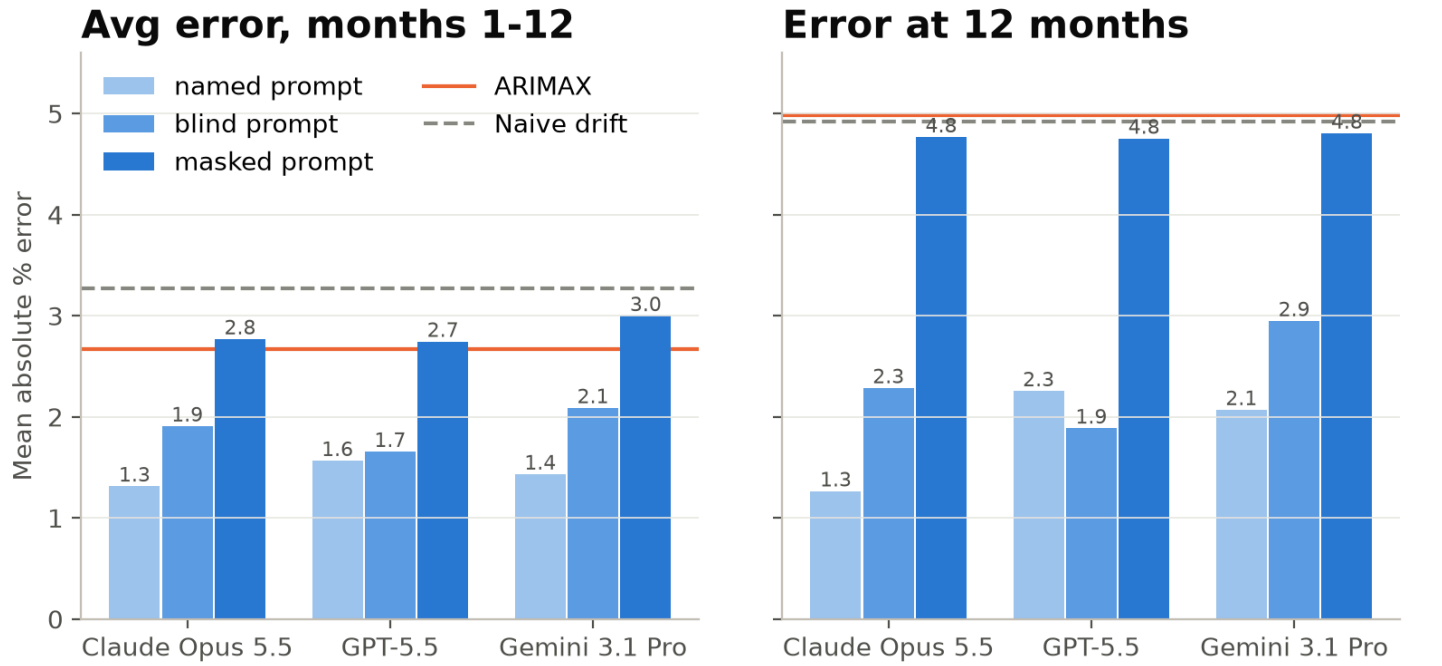


Same model, two tests

1st → 10th

1st in the 64-month test (2.2% error), 10th in repeated 12-month tests. LightGBM's lead did not hold under our main evaluation.

An alternative tool: AI's edge faded once we removed the clues



Claude: named → masked

1.3 → 2.8%

Claude's average error rose from 1.3% to 2.8%, close to ARIMAX's 2.67%. These historical periods may already be familiar to the AI.

Result 3 · AI as an alternative, how we tested

Masking cut recognition, but didn't remove it

Named

LA, real dates, dollars, rates

Most clues

Blind

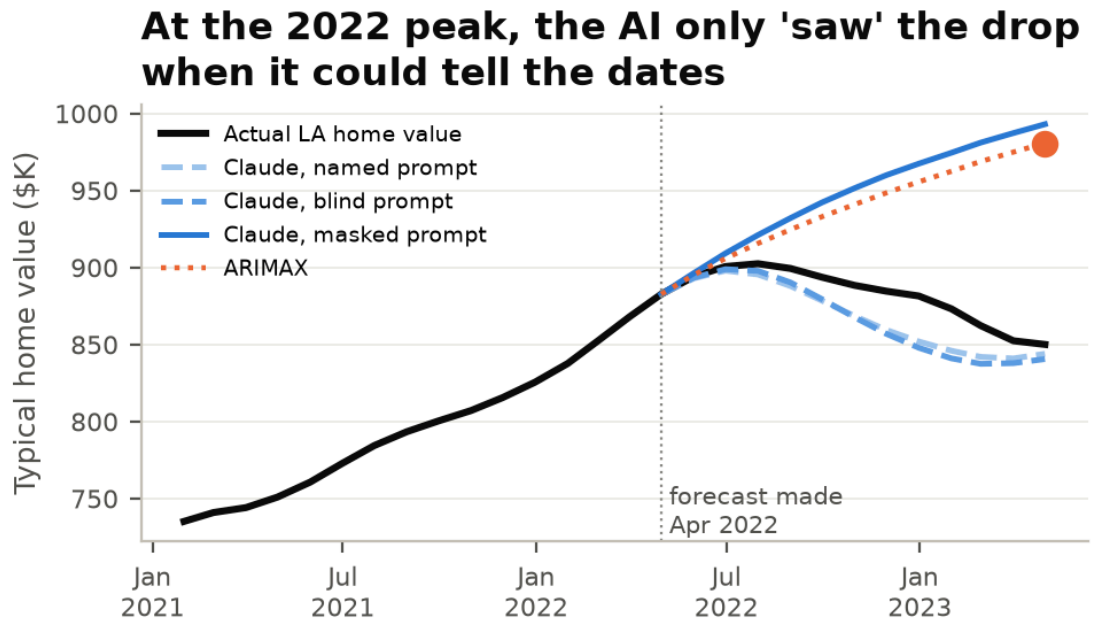
Place, dates hidden; rates kept

Rates can reveal dates

Masked

10 years, rescaled; no place, dates, rates

Fewest clues

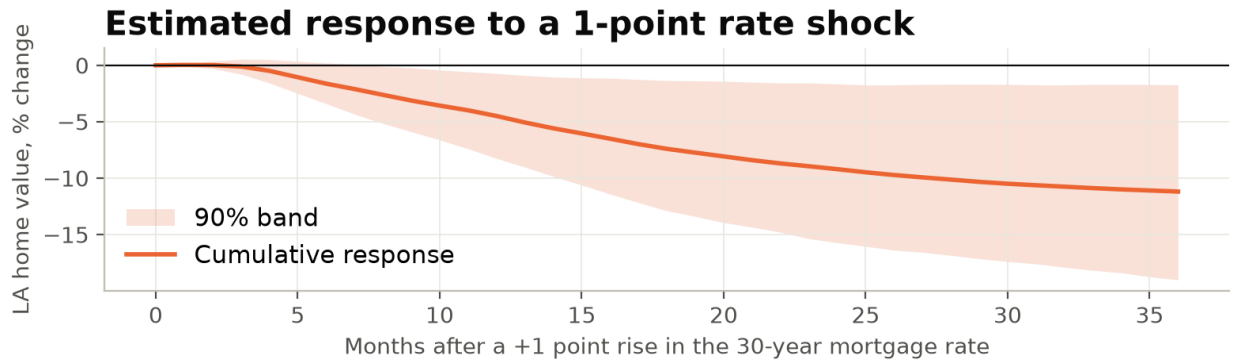


Name this series Claude named LA in 16/18 blind and 0/18 masked prompts, yet dated masked data to about a month. Given 26 years, not 10, it called the 2022 fall (2.47% error).

Result 4 · Mortgage rates and outlook

Rates act with a delay; we forecast +4.8%

Estimated home-value response to a 1-point mortgage-rate shock



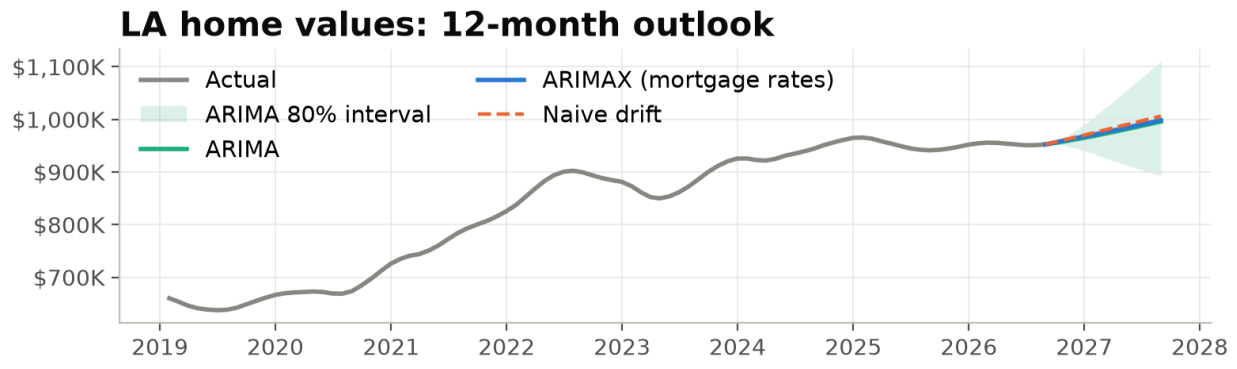
−9.2%

VAR-estimated response after two years (−4.5% after one)

p = 0.052

Granger: borderline. ARIMAX: rate p = 0.16, jobs p = 0.29

Our forecast: about 5% higher by August 2027, with a wide range



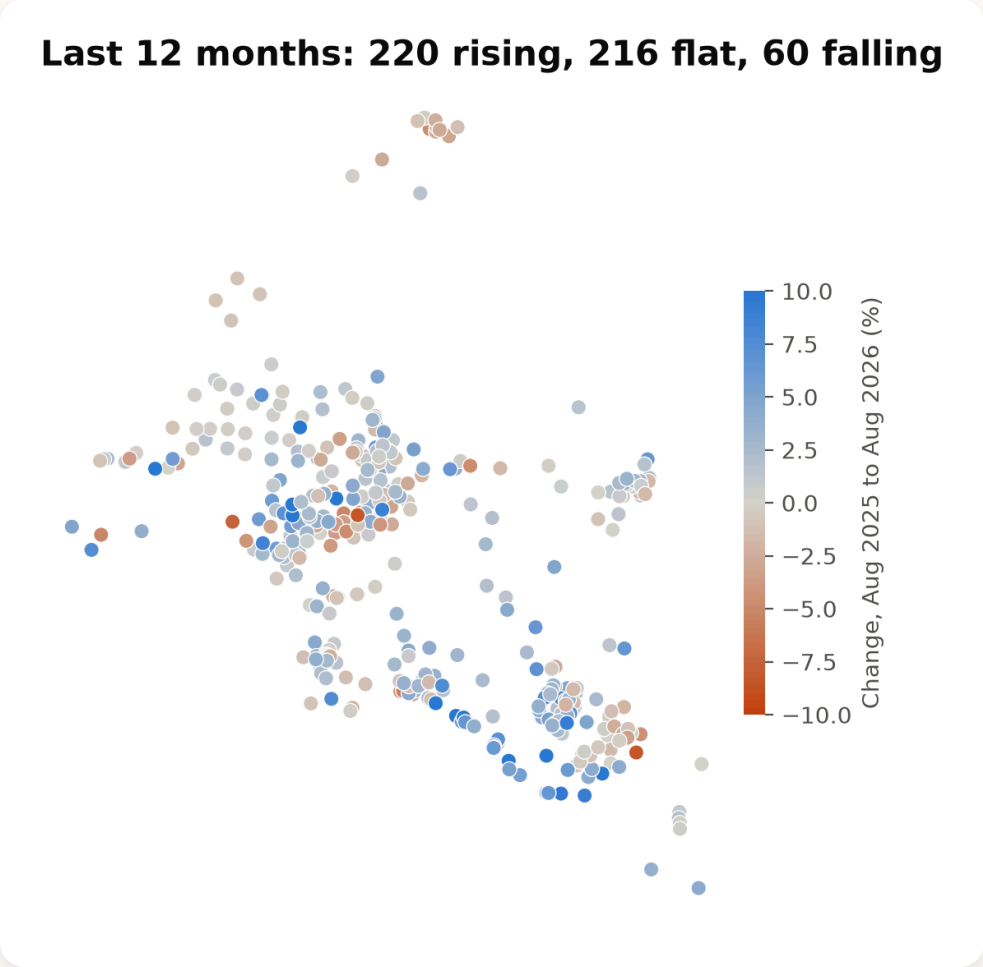
$998K (+4.8%)

ARIMAX forecast, August 2027

−6% to +17%

for context: ARIMA's 80% interval by August 2027

Some neighborhoods are rising while others are falling



220 / 216 / 60

rising >2% · within ±2% · falling >2%, last 12 months (496 neighborhoods with recent data)

152 of 496

still below their 2022 peak, mostly in LA County

−1.2 pts

in LA County, doubling the starting home value went with a 1.2-point larger decline after the 2022 peak

Live map: neiltd.github.io/la-housing-forecast

Takeaways

Our answer: simple models held up best

01

Simple models were competitive

ARIMAX had the lowest average error; naive drift was equally competitive at month 12.

02

Test AI on outcomes it cannot know

Historical AI tests can be affected by recognition of past events.

03

Mortgage rates matter, slowly

Jobs, income and crime didn't improve the forecast; turning points stayed hard to call.

04

The metro average is only part of the picture

Results varied; per-neighborhood ARIMA beat naive drift (2.8% vs 3.5%).

Next steps Test more cities; score the AI live on months after its training cutoff.

What each method knew at each forecast date

Method	Price history	Real dates	Place	Mortgage rate	Jobs
ARIMA, naive drift, ML models	2000 to forecast date	no	no	no	no
ARIMAX	2000 to forecast date	no	no	**yes, lag 24**	no
ARIMAX + jobs	2000 to forecast date	no	no	**yes, lag 24**	**yes, lag 12**
AI, named prompt	2000 to forecast date	**yes**	**yes**	**yes**	no
AI, blind prompt	2000 to forecast date	no	no	**yes**	no
AI, masked prompt	last 10 years	no	no	no	no
AI, masked, all history (Claude only)	2000 to forecast date	no	no	no	no

The 14 models in one line each

Model	Family	In one line
Naive drift	Baseline	Last value, grown at the average monthly growth since 2000 (a random walk with drift).
ARIMA	Classical	Forecasts from the series' own recent values and recent surprises, after differencing once; ARIMA(1,1,2) on log price.
Holt (damped)	Classical	Exponential smoothing of the level and trend, with the trend slowly flattening out.
Prophet	Classical	Meta's curve-fitting model: one smooth trend line plus changepoints; does not start from the latest value.
Prophet + COVID & rates	Classical	Prophet with the 2020-22 boom marked as a one-off and the mortgage rate (lagged 24 months) added.
XGBoost	ML	Boosted decision trees predicting next month's growth from the last 12 months of growth.
LightGBM	ML	A faster boosted-tree method, same inputs as XGBoost.
LSTM	Deep learning	A recurrent neural network that reads the last 24 months of growth and predicts the next.
GRU	Deep learning	A simpler recurrent network, same setup as the LSTM.
Chronos (foundation model)	AI, pretrained	Amazon's pretrained time-series model, used as is: it reads the price history and forecasts, never trained on our data.
ARIMAX	Outside info	ARIMA on monthly growth plus the 12-month change in the mortgage rate, lagged 24 months.
ARIMAX + jobs	Outside info	ARIMAX plus 12-month payroll job growth in LA and Orange County, lagged 12 months.
VAR (growth + rate)	Outside info	Home growth and rate changes forecast each other (11 monthly lags).
Factor VAR (neighborhoods)	Outside info	Summarizes ~400 neighborhoods into 2 common factors and forecasts them together with metro growth.

Appendix A3 · Rolling-origin test

Full rolling-origin results, 14 models

Model	1 mo	3 mo	6 mo	12 mo	Avg 1-12	RMSE $	RMSE rank
ARIMAX	**0.09**	**0.81**	**2.45**	**4.98**	**2.67**	**35.8K**	**1**
ARIMA	0.09	0.78	2.53	5.37	2.77	38.1K	4
ARIMAX + jobs	0.12	0.94	2.63	5.12	2.81	36.2K	2
VAR (growth + rate)	0.14	0.94	2.56	5.74	2.93	47.6K	10
LSTM	0.12	0.89	2.95	5.58	3.02	43.3K	6
GRU	0.11	0.93	2.99	5.86	3.18	41.5K	5
Naive drift	0.69	1.93	3.48	4.92	3.27	37.6K	3
Holt (damped)	0.28	1.15	3.04	6.34	3.39	44.3K	7
Chronos (foundation model)	0.69	1.61	3.44	6.13	3.56	45.3K	8
LightGBM	0.22	1.23	3.32	6.97	3.60	46.5K	9
Factor VAR (neighborhoods)	0.16	1.15	3.56	6.60	3.64	50.7K	12
XGBoost	0.25	1.25	3.40	6.82	3.65	47.8K	11
Prophet + COVID & rates	3.27	4.08	5.32	6.45	5.12	53.5K	13
Prophet	5.00	5.76	6.73	8.62	6.89	70.2K	14

Memory checks on Claude

Check (Claude only)	Result
Name the blind series	Los Angeles 16 of 18, California 17 of 18; dates exact every time
Name the masked series	Los Angeles 0 of 18; guesses: unknown 7, Seattle 6, Portland 2, San Francisco 1, Dallas 1, Denver 1
Date the masked series	median 1 month off, max 4
Masked, all history since 2000	2.47% avg error (ARIMAX 2.67%), DM p = 0.25; April 2022 forecast -4.7% (actual -3.7%)
Earlier run through Claude Code	named 1.58%, masked 2.67%; the clean API rerun changed little

Appendix A5 · Trend check

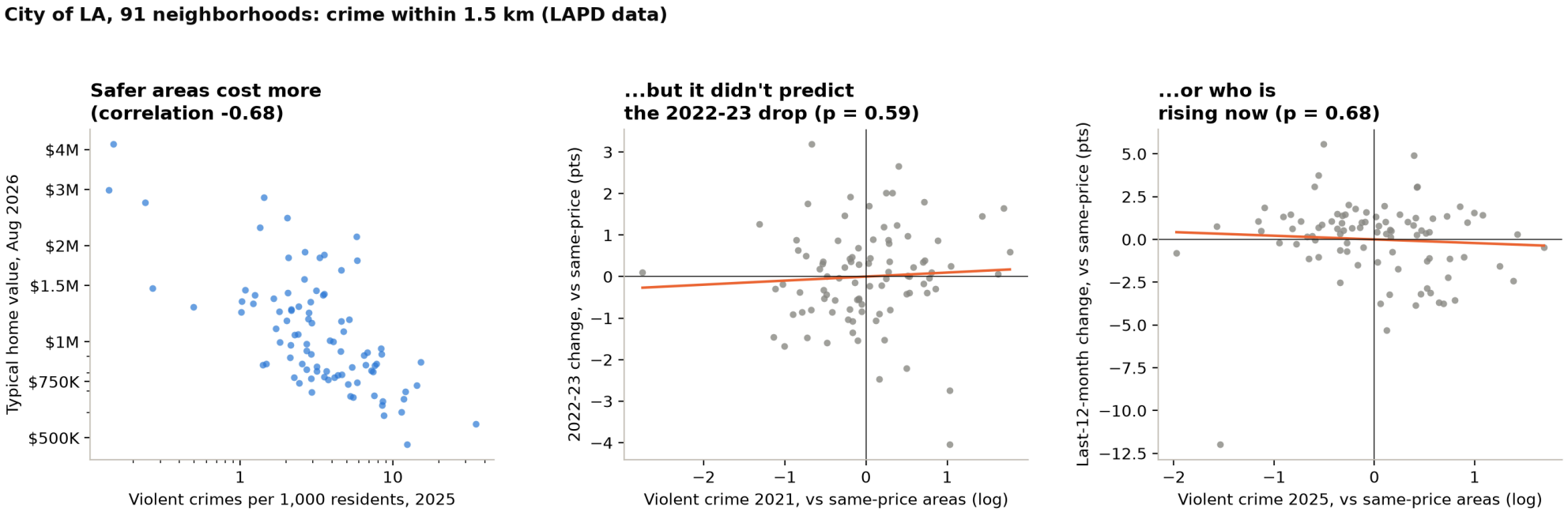
Named and blind AI tracked what happened; masked AI tracked the trend

Prompt	Claude	GPT	Gemini	Follows
Named	0.97 / 0.14	0.91 / 0.16	0.90 / 0.28	what happened
Blind	0.89 / 0.19	0.94 / 0.23	0.83 / 0.15	what happened
Masked	0.21 / 0.83	0.26 / 0.87	0.15 / 0.49	recent trend
ARIMAX	0.37 / 0.66			recent trend

Note Each cell: correlation of the 12-month forecast with the actual 12-month change / with the last 3 months' growth, across 18 forecast dates.

Appendix A6 · Only if asked about crime

Crime tracks price, but it didn't predict the 2022 drop or today's winners



Why it's in the appendix Crime data covers only the City of LA (91 of 496 neighborhoods), and it adds nothing to price prediction: at the same price level, p = 0.59 and 0.68. Price already carries it.